

# TrendScope

## Offline-First News Reader

Android News Aggregator with RSS, Translation,  
Bookmarks & Offline Support

Kotlin

Room DB

Coroutines

Material 3

Team Lead Review — July 2026

## PROBLEM STATEMENT

# Challenge

Users need a single app that aggregates diverse news sources, supports Indian regional languages, works offline, and lets them save & bookmark articles — all with a clean, modern UI.

## ARCHITECTURE OVERVIEW

## Tech Stack

- ▶ Kotlin + Room (SQLite)
- ▶ Coroutines + Dispatchers
- ▶ MVVM-like separation
- ▶ Activities ↔ DAO ↔ RSS Parser
- ▶ Material Design 3
- ▶ DayNight theming
- ▶ R8 minification
- ▶ ViewBinding

**Key patterns:** Parallel RSS fetch (Semaphore 8), OkHttp image caching, OnConflictStrategy.IGNORE for bookmark safety

## KEY FEATURE

# Multi-Source RSS Aggregation

- ▶ 100+ RSS feeds across Indian news categories (Tech, Business, Sports, etc.)
- ▶ Atom + RSS 2.0 parsing with per-feed 25s timeout
- ▶ Concurrent fetch capped at 8 simultaneous feeds via Semaphore
- ▶ Self-heals: 100ms retry on failure, individual feed errors don't block others
- ▶ Deduplication by article link; max 200 articles per refresh

## KEY FEATURE

# Translation Engine

- ▶ 22 Indian languages supported (Hindi, Bengali, Tamil, Telugu, etc.)
- ▶ Batch translation with 400ms delays to avoid rate limiting
- ▶ Per-article cached translations stored in Room
- ▶ Source-language auto-detection; user picks any target language
- ▶ Swap source ↔ target with one tap
- ▶ Progress tracking: "Translating (3/15)..."

## KEY FEATURE

## Offline News Download

- ▶ One-tap download caches **article + image** locally
- ▶ ImageCacheManager stores images as MD5-hashed JPEGs in `filesDir/offline_images/`
- ▶ Dedicated **Offline tab** (tab 2) lists all saved articles
- ▶ Offline banner auto-shows when ConnectivityManager reports no internet
- ▶ Delete with confirmation dialog frees storage
- ▶ Search works on offline articles too

## KEY FEATURE

# Bookmarks

- ▶ Bookmarks persisted via Room with `OnConflictStrategy.IGNORE` — never clobbered by RSS refresh
- ▶ `toggleBookmark()` uses single atomic `insertArticle()` — no TOCTOU race condition
- ▶ Dedicated **Bookmarks tab** (tab 1)
- ▶ Star icon (filled/empty) on every article detail screen

## KEY FEATURE

# Search & Filters

- ▶ Full-text search across all cached articles
- ▶ Advanced Search dialog: keyword, category filter, date range, newest/oldest sort
- ▶ Search history persisted (last 20 queries)
- ▶ Filter chips for quick refinement



UI / UX

## Dark Mode & UI Details

- ▶ Light / Dark toggle in toolbar overflow menu — persisted in SharedPreferences
- ▶ DayNight.NoActionBar theme with custom values-night/colors.xml
- ▶ Breaking-news badge (articles <60 min old)
- ▶ Day-based section headers (Today / Yesterday / This Week / This Month / Earlier)
- ▶ Favicon per news source, author display, share button

## LOOKING AHEAD

# Roadmap & Next Steps

Done

Planned

- ▶ ✓ RSS aggregation (100+ feeds)
- ▶ ✓ 22-language translation
- ▶ ✓ Offline download with image caching
- ▶ ✓ Bookmarks, search, dark mode
- ▶ ☐ Push notifications for breaking news
- ▶ ☐ Background sync via WorkManager
- ▶ ☐ Bookmark folders / categories
- ▶ ☐ Tablet / landscape layout

~4K LOC · Single module · Clean Room schema · Passes R8 minification